NH5 Highway AI

Technical report - v8 model, conversion, pipeline and website

28 September 2026

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1. System overview

Stage	What it does
Detection	YOLOv8n, 12 classes, 960 px input, ByteTrack tracking
Events	Temporal rules for accident / fire / road-work alerts; zone and lane rules with a road map
Plates	Plate crop -> clarity filter -> PaddleOCR -> Indian-format check -> multi-frame voting
Website	Upload, live camera, Library, Excel report
Deployment	FP32 and INT8 ONNX for the Rubik Pi

Classes (id): 0 person, 1 bicycle, 2 car, 3 motorcycle, 4 bus, 5 truck, 6 pothole, 7 license_plate, 8 auto, 9 road_construction, 10 accident_scene, 11 fire_accident. Rules: riders are part of the vehicle box (person = pedestrians only); fire and smoke form one fire_accident box; fire_accident only when a vehicle is burning; a lone cone/sign is not road work.

2. Dataset (v8)

Built from public and project sources (COCO-style traffic, Indian road datasets, CCPD/plate sets, RDD2022 potholes (D40 only), road-work sets, accident and fire sets, night scenes). Cleaned in place on 24 Sep: ~170k missing person/vehicle boxes added (78,614 cars), road-work zones rebuilt from object annotations, fire restricted to vehicle fires, 5,328 noisy images removed. The vedios/ folder is test-only and never used for training.

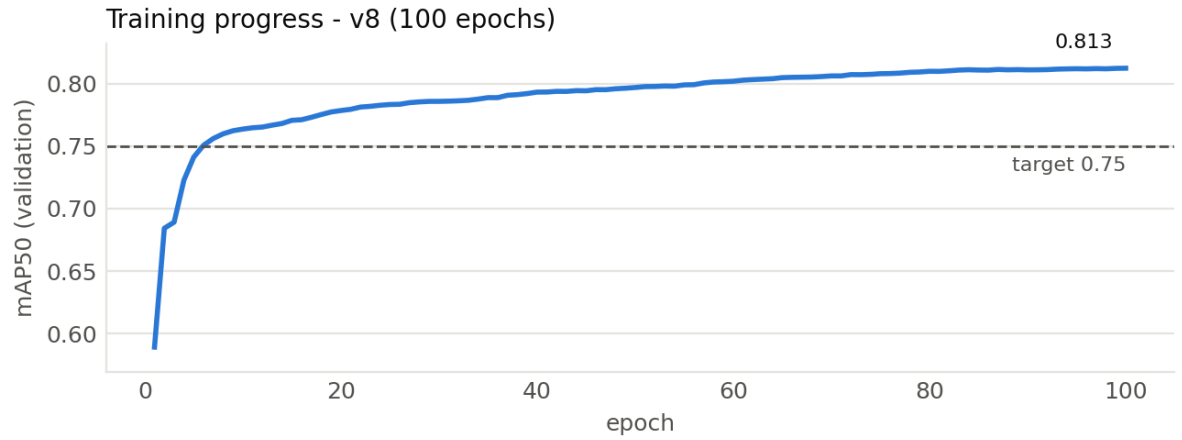
Class	Train boxes	Val boxes	Test boxes
person	81,869	12,140	11,518
bicycle	14,827	601	572
car	167,200	14,704	13,224
motorcycle	48,801	5,493	3,225
bus	23,578	1,652	1,552
truck	38,289	4,159	3,491
pothole	27,479	1,409	802
license_plate	17,524	2,000	2,062
auto	33,903	2,957	2,286
road_construction	11,414	1,484	1,329
accident_scene	14,038	971	942
fire_accident	3,452	549	602
Images	**88,083**	**8,819**	**8,040**

Balancing: fire images are repeated 5x and pothole / bicycle / road-work images 2x in each epoch (128,530 samples per epoch from 82,975 unique images).

3. Training

Setting	Value	Setting	Value
Model	YOLOv8n (fine-tuned)	Image size	960
Epochs	100	Batch	32
Optimizer	auto (SGD), lr0 0.01, lrf 0.01, cosine	Warm-up	3 epochs
Augmentation	mosaic 1.0 (off last 10 epochs), hsv_v 0.4	Patience	100 (all epochs run)
Hardware	RTX 3090 Ti 24 GB, 62 GB RAM	Time	~16.5 min/epoch, finished 26 Sep 15:10

Epoch	21	40	50	60	70	80	90	100
mAP50 (val)	0.780	0.793	0.797	0.802	0.806	0.810	0.811	0.813
mAP50-95	0.570	0.584	0.588	0.594	0.598	0.601	0.601	0.602

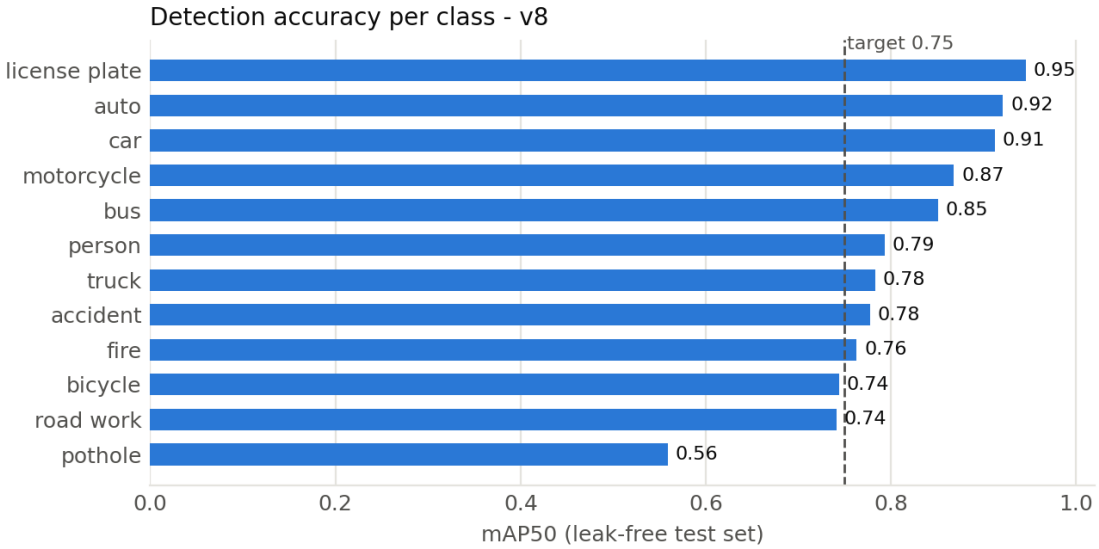


4. Evaluation

Per-class mAP50. Full test = 8,040 images. **Leak-free test** = 6,786 test images with no near-duplicate (dHash distance <= 4) in train; about 16% of val/test had a near-copy in train, mainly consecutive video frames.

Class	Full test (.pt)	Leak-free test (.pt)	Full test (INT8)
license_plate	0.939	0.946	0.935
auto	0.921	0.921	0.915
car	0.916	0.912	0.911
motorcycle	0.871	0.868	0.864
bus	0.851	0.851	0.850
fire_accident	0.837	0.763	0.804
accident_scene	0.802	0.777	0.772
person	0.801	0.793	0.794
truck	0.792	0.783	0.785
road_construction	0.745	0.741	0.716
bicycle	0.740	0.744	0.721
pothole	0.597	0.559	0.590
all	**0.818 / 0.606**	**0.805 / 0.590**	**0.805 / 0.589**

Values in the last row are mAP50 / mAP50-95.



Video tests (14 test videos, full pipeline incl. PaddleOCR)

Model	Vehicle boxes	Valid plates	Accident alert frames	Fire alert frames	Flicker %	Jitter %
v5	104,630	10	0 (no class)	0 (no class)	29.8	12.1
v6	43,023	31	1,514	0 (no class)	44.4	24.0
v8 final	133,439	31	1,161	438	35.8	8.3

Snapshots (epochs 30-90) were tested the same way; the final model is best on 10 of 11 classes. The epoch-60 checkpoint reads more plates from overhead cameras and is kept as an alternative. Weight averaging of checkpoints was tried and rejected (mAP50 0.765-0.775).

5. INT8 ONNX conversion

- Step 1: export best.pt to FP32 ONNX (opset 17, 1x3x960x960, no NMS, onnxslim). Verified against PyTorch: max box diff 0.03 px, score diff 1e-5.
- Step 2: static calibration with ONNX Runtime, QDQ format, weights INT8 per-channel symmetric, activations UINT8. 600 calibration images from TRAIN (all 12 classes + night). The 24 Detect-head decode nodes stay FP32 (the output mixes 0-960 px boxes with 0-1 scores).
- Methods compared on the test set (mAP50): Percentile 99.999 = **0.805** (chosen), Entropy 0.804, MinMax 0.782.
- Speed on PC CPU (4 threads): FP32 53 ms, INT8 26 ms per frame. Output 1x16x18900 (cx, cy, w, h + 12 class scores); NMS is done by the application.
- Pitfalls found: Entropy/Percentile calibration must be fed in chunks (~500 MB of activations per 960 image); ORT 1.23 CalibMaxIntermediateOutputs discards MinMax batches ("No data is collected").

6. Number-plate reading

- Indian formats only: state code (incl. TG, BH series), 1-2 digit district (1 digit only for DL), 1-3 series letters, 4 digits; at most 2 look-alike characters corrected (O/0, S/5, B/8 ...); anything else rejected.
- Clarity filter per crop: upscale to 128 px, non-local-means denoise, deskew (2-20 deg), CLAHE, unsharp mask.
- Two-line (bike/auto) plates: split at the emptiest row and read top line first.
- Video test: per video, a plate is confirmed when >= 2 frames agree, mean OCR conf >= 0.85 and best >= 0.93; readings within 2 characters are merged; rejects logged with the reason.
- Website: OCR conf >= 0.85, 5 agreeing reads per track, near-duplicate merge, cheapest-first OCR with early exit and no OCR after confirmation (1.7-2x faster, same plates).
- Result on the 14 test videos: 11 confirmed plates, all correct, 0 false (previously 83 readings incl. UK and junk).

7. Website

Area	Behaviour / threshold
Model	v8 detection model with PaddleOCR for number plates
Fire alert	fire_accident conf >= 0.45 in 60% of a 1 s window; box must contain >= 1.2% orange/red flame pixels
Accident alert	accident_scene conf >= 0.50 in 60% of a 1 s window; weaker boxes not drawn
Cooldown	30 s between repeated accident / fire / road-work alerts
Vehicle counts	unique tracks per type (car, motorcycle, auto, bus, truck, bicycle); people separate
Plate counts	one record per plate per video, 'Times Seen' + 'Last Seen' in Excel, xN on pages
Speed display	upload progress (MB/s); frames X/Y, fps, ms/frame, time left; live page too
Known gaps	images cannot raise alerts (1 s rule); concurrent uploads share one tracker

Results per test video (v8, full video test)

Video	Frames	Vehicle boxes	Person boxes	Accident alert frames	Fire alert frames	Road-work boxes	Jitter %
AQNm... (short clip)	866	1,630	1	0	0	0	13.4
Create_a_highly_realistic	240	2,830	334	0	0	0	6.6
License Plate Test (UK)	1,800	36,947	1	0	0	0	3.1
MicrosoftTeams (fallen bike)	287	218	967	18	0	0	6.9
MicrosoftTeams (1)	888	16,918	336	246	0	0	3.4
MicrosoftTeams (2)	545	8,037	1,642	0	0	0	5.2
VID20260824132342	1,844	11,953	471	0	0	57	7.1
VID20260824132956	2,110	10,074	3,471	0	0	20	8.5
VID20260824133121	866	13,494	3,012	0	0	0	4.7
WhatsApp 12.25.40 (car fire)	304	271	4	0	111	4	26.6
WhatsApp 12.26.27	1,916	6,203	108	0	0	0	7.0
real_accident_cctv_test	2,298	14,453	1,909	897	148	0	3.5
reference_vedio	1,949	9,868	7,627	0	0	0	8.2
videoplayback (4) (car fire)	304	543	9	0	179	0	11.5

Jitter = average frame-to-frame change of the object count (lower = steadier). The 148 fire alert frames on the CCTV compilation are the dust cloud of a truck crash; the website now blocks this with the flame-colour check.

INT8 calibration comparison (test set, 8,040 images)

Model	Precision	Recall	mAP50	mAP50-95	Size	CPU ms/frame (4 threads)
.pt (PyTorch)	0.785	0.762	0.818	0.606	6 MB	-
FP32 ONNX	0.784	0.760	0.814	0.600	12 MB	53
INT8 Percentile 99.999 (chosen)	0.761	0.765	**0.805**	0.589	3.7 MB	26
INT8 Entropy	0.765	0.761	0.804	0.586	3.7 MB	-
INT8 MinMax	0.783	0.705	0.782	0.567	3.7 MB	26

Class drops FP32 -> INT8 (mAP50): fire -2.6, road work -2.3, bicycle -1.6, pothole -1.3, all other classes < 1 point; license_plate unchanged (0.935).

8. Known issues

#	Issue	Evidence	Fix
1	Potholes weak	mAP50 0.56-0.60, recall ~0.45	label clean-up
2	Accident fires 1-3 s before impact	CCTV clips; ~1 in 5 accident training frames are pre-impact	relabel pre-impact frames
3	Accident on close-up normal cars	0.34-0.68 conf	website 0.50 rule; more negatives
4	Road construction false detections	data added; bollards / lone signs detected (0.48 conf)	train separately
5	Overhead plates	125 -> 1 plate crops (ep30 -> ep100)	more data
6	Aftermath photos missed	0.11 / 0 conf on test photos	more data
7	Smoke-only fire	no flames -> no alert	more data
8	Two-line plates	bike / auto plates sometimes read with one character different	more labels
9	Concurrent uploads	same video 11/16/47 vehicles	code fix
10	Image uploads cannot alert	fire 0.89 on a photo, no alert	code fix
11	INT8 drop	fire -2.6, road work -2.3 points	calibration
12	Rubik Pi speed unknown	PC only	board test

Note - road construction: road-construction data has been added to the dataset and the model detects work zones, but it still gives false detections (for example permanent roadside bollards and lone signs). Road construction **will be trained separately** as its own improvement step.

Note - lane change and wrong-direction detection: these will be checked once the camera is fixed in the position where the model needs to be tested, because both depend on the camera view (the lanes and the allowed direction of travel are set for that fixed position).